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10 January 1961
SUPERSEDING
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18 February 1958

MILITARY SPECIFICATION

BOILERS, HEATING, LOW PRESSURE, HOT WATER NAVAL SHIPBOARD

1. SCOPE

1.1 Scope. - This specification covers oil fired, straight fire, tube type hot water heating boilers for shipboard use.

1.2 Classification. - The boilers shall be of the following classes, styles, and sizes, as specified (see 6.1):

Class I - Carbon steel construction.
Class II - Nonmagnetic construction.
Style A - Horizontal design.
Style B - Vertical design.
Size 1 - 150,000 British thermal units (B.t.u.)/hour capacity.
Size 2 - 250,000 B.t.u./hour capacity.
Size 3 - 350,000 B.t.u./hour capacity.
Size 4 - 450,000 B.t.u./hour capacity.

1.2.1 Capacity of the boiler is defined as quantity of heat, in B.t.u./hour transferred into water forced circulated through the boiler, as measured at the boiler hot water outlet, when the boiler is burning Diesel fuel oil conforming to Specification MIL-F-16884.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawing, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-766 - Steel Plates, Sheets, and Strip -- Corrosion Resisting.

MILITARY

MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-B-233 - Boxes, Repair Parts.

MILITARY (Cont'd)

MIL-D-963 - Drawings, Production, Procedure for Procurement of.
MIL-D-5480 - Data, Engineering and Technical: Reproduction Requirements for.
MIL-P-15024 - Plates, Identification Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).
MIL-P-15137 - Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-I-16165 - Interference Shielding, Electrical Systems.
MIL-P-16298 - Preservation, Packaging, Packing, and Marking of Electric Machine Having Rotating Parts and Associated Repair Parts.
MIL-F-16884 - Fuel Oil, Diesel, Marine.
MIL-I-16910 - Interference Measurement, Radio, Methods and Limits; 14 Kilocycles to 1000 Megacycles.
MIL-M-17059 - Motors, Alternating-Current, Fractional HP (Shipboard Use).
MIL-I-17166 - Iron Castings; Nodular Graphitic (Ductile Iron) (for Shipboard Applications).
MIL-P-17286 - Propulsion and Auxiliary Steam Turbines and Gears (Including Repair Parts, Tools, Accessories

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and Instruments); Packaging of.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-M-17556 - Motor, Direct-Current, Fractional HP (Shipboard Use).
MIL-D-70327 - Drawings, Engineering and Associated Lists.

STANDARDS

MILITARY

MIL-STD-22 - Welded-Joint Designs.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.

DRAWING

BUREAU OF SHIPS

RE-49A501 - Coupling, Spherical Seat Design Data.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

SB-111 - Copper and Copper-Alloy Seamless Condenser Tubes and Ferrule Stock.
SB-171 - Copper-Alloy Condenser Tube Plates.

Boiler Construction Code.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 29 West Thirty-Ninth Street, New York, N.Y.).

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.).

3. REQUIREMENTS

3.1 Preproduction model. - One boiler shall be submitted to a laboratory satisfactory to the bureau or agency concerned for the preproduction inspection specified in 4.2.1.

3.2 Electrical characteristics. - The boiler shall be designed to operate on the electric power supply of one of the following ratings, as specified (see 6.1):

Direct current (d.c.)	Alternating current (a.c.)		
Volts	Volts	Phase	Cycle
115	440	3	60
230	220	3	60
Special	110	3	60
	230	Single	--
	115	Single	--
	Special	---	--

3.3 Material. -

3.3.1 Cast iron shall not be used in any part of the boiler. Nodular iron conforming to Specification MIL-I-17166 may be used in boiler auxiliaries, accessories, and controls, where no other substitute for cast iron is practicable, provided approval of the specific application is obtained from the bureau or agency concerned.

3.3.2 Class I - carbon steel construction. - Except as otherwise specified herein, material used in construction of class I boilers shall conform to the A.S.M.E. Boiler Construction Code.

3.3.3 Class II - nonmagnetic construction. -

3.3.3.1 General requirements. - Boilers specified herein are to be installed on ships whose magnetic signature must be extremely low. Therefore great importance is attached to use of nonmagnetic materials wherever possible without involving development or affecting the design characteristics. Where experience with working parts mandates the use of magnetic ferrous materials in the absence of suitable nonmagnetic substitutes, such parts should be as widely dispersed as practicable. On the basis of this consideration, the use of ferrous materials in crankshafts, piston rings, cylinder liners, threaded parts, and ball bearings will be considered by the bureau or agency concerned if the contractor considers there is no suitable nonmagnetic substitute. Other parts of all components used in the boiler assembly shall be manufactured from

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nonmagnetic materials. Except as otherwise specified herein, material used in the construction of equipment shall be considered nonmagnetic if it has a permeability of less than 1.08 before fabrication and not greater than 2.0 after fabrication. Material used shall be fully identified on preliminary drawings (see 6.2) and on all drawings submitted by the contractor for approval after award of the contract or order (see 3.10).

3.3.3.2 Materials. - The following materials shall be used in the boiler shell assembly (including tubes):

(a) 70-30 copper-nickel alloy plate in accordance with A.S.M.E. Publication SB-171 for shell assembly proper and all pressure parts.
(b) 90-10 copper-nickel alloy seamless tubes (annealed) in accordance with A.S.M.E. Publication SB-111 for tubes.
(c) Steel, corrosion-resisting conforming to classes 304, 321 or 347, low permeability, of Specification QQ-S-766 may be used for such applications as breechings, cover plates, structural and support members, provided these members do not form an integral part of the pressure vessel and provided they are not welded or brazed to the pressure vessel. Bronze, brass or aluminum may also be used for supports and for jacket over the shell insulation.
(d) Bronze or brass for valves, pipe fittings and piping incident to the boiler assembly, provided that the particular material is suitable for the purpose intended.

3.3.3.3 Electrical equipment. -

(a) All items of electrical equipment shall be fabricated of nonmagnetic material, except for the following parts:
  (1) Parts required in the magnetic circuit of the particular item of equipment.
  (2) Structural parts of motors, rotary converters, contactors, relays, thermostats, and pressure switches which would entail extensive modification and redesign to permit substitution of nonmagnetic material.
(b) Electric controllers, solenoids, and switches shall have enclosure and panels of suitable nonmagnetic metals.
(c) Stray magnetic fields caused by electrical circuits shall be kept to a minimum by making the area of electric current loops as small as possible and arranging adjacent loops so that the fields will be in opposition. Current loops shall enclose no magnetic material unless essential for proper performance of equipment.
(d) In a.c. motors, all connections between stator winding connections shall be made as short as possible. The power leads to the stator winding shall be run through the same opening in the motor frame.

3.4 Radio interference suppression. -

3.4.1 The boiler automatic high tension ignition and combustion control system shall be arranged to minimize radio interference, both radiated and conducted, through the use of suitable enclosures, shielding conduits, connectors, and line filters, as required. External leads shall be shielded to reduce radio interference by shielded metallic conduit of proper diameter with spherical-seat ball and cone fittings in accordance with Specification MIL-I-16165 and Drawing RE-49A501, as applicable. For the foregoing type of ignition, and any other type of ignition which may be used, all electrical connections and wiring shall be waterproof. Measurements and limits specified in Specification MIL-I-16910 shall be used as a guide in designing radio interference suppression. The arrangement of radio interference suppression and all equipment used for same shall be subject to approval by the bureau or agency concerned.

3.5 Design. -

3.5.1 Except as otherwise specified herein, the boiler shall be designed and manufactured in accordance with the Boiler Construction Code.

3.5.2 The boiler shall be of the following design and performance characteristics:

Pressure design, pounds per square inch (psi)	30.
Pressure operating, psi	15.
Temperature, operating	
burner cut-in	175° F. approximately.
burner cut-off	195° F. approximately.
Temperature, air	100° F.
Boiler efficiency	75 percent minimum.
Percent carbon dioxide at the boiler smoke outlet	10 percent minimum.
Gas temperature at the boiler smoke outlet	600° F. maximum.

3.5.3 The boiler assembly shall include the following components:

(a) Burner.
(b) Forced draft blower.
(c) Fuel oil pump.
(d) Motors and controllers as required by driven auxiliaries.
(e) Automatic ignition and combustion control system, including the following:

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(1) Rotary inverter and ignition transformer as necessary to supply alternating electric current for ignition (on d. c. powered boilers).
(2) Thermostatic combustion control.
(3) Safety combustion control and relay (stack switch).
(4) Fuel oil solenoid valve.
(5) Control circuit transformer (on a. c. powered boilers).
(f) Oil pressure regulating and relief valve.
(g) Safety relief valve.
(h) Bottom blow-off valve.
(i) Pressure and altitude gage and thermometer.
(j) Fuel oil pressure gage.
(k) Fuel oil strainer.
(l) Combustion observation window.
(m) Cleanout openings and piping connections as necessary.
(n) Fuel oil filter.

3. 5. 4 The boiler shall be equipped with necessary brickwork, insulation, and lagging consisting of a metal jacket over insulation.

3. 5. 5 Essentially, the boiler shall be a self-contained and complete unit when connected with electrical, fuel oil, and water supply and return lines to be furnished at the place of installation. All parts of the boiler assembly shall be arranged with respect to each other to permit easy access for servicing or repair and replacement of parts.

3. 5. 6 The boiler design shall permit normal operation of the boiler when subject to inclination to an angle up to 45 degrees in any direction. Provisions shall be made for firmly securing the boiler to deck or foundation.

3. 5. 7 The boiler shall be of either style A, horizontal or style B, vertical (see 6. 1) straight tube design. The weight and physical dimensions of the boiler complete with insulation and lagging and all auxiliaries, accessories, and controls shall be the minimum commensurate with reliable and efficient operation, and shall not exceed the limitations specified in table I:

Table I - Limitations.

Size	Style A-horizontal			Style B-vertical			Weight dry
	Overall			Overall			
	Length	Width	Height	Length	Width	Height	
	Inches	Inches	Inches	Inches	Inches	Inches	Pounds
1	36	21	36	30	21	51	600
2	42	24	42	33	24	54	700
3	48	27	45	39	27	61	850
4	54	30	48	42	30	64	1000

3. 5. 8 The finished diameters of tube holes in tube sheet shall not exceed the following:

Size tube Inches	Tube hole diameter Inches
1	1. 02
1-1/4	1. 28
1-1/2	1. 53
1-3/4	1. 78
2	2. 03
2-1/4	2. 28
2-1/2	2. 53

3. 5. 9 The sharp edge of tube holes shall be taken off on both sides of the plate with a file or other tool.

3. 5. 10 The boiler shall be of welded construction. Welding shall be in accordance with Standard MIL-STD-278.

3. 5. 11 Longitudinal seams of shells shall be welded from inside and outside. Joints shall comply with joint numbers B-2, B-5, B-14, or B-31 of Standard MIL-STD-22.

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3.5.12 Plain unflanged tube sheets shall be attached to shells and combustion chambers by means of a single "J" groove joint, fillet reinforced weld, similar to joint number T-42 of Standard MIL-STD-22, except that the root gap shall not exceed 1/8 inch. The size of the reinforcing fillet shall not be less than the thickness of the thinner of the two parts joined.

3.5.13 The following methods shall be used for connecting tubes to the tube sheets: Tubes shall be rolled, beaded, and welded around the edge of the bead.

3.5.13.1 The minimum allowable thickness of noncylindrical shells shall be in accordance with applicable paragraph of the A.S.M.E. Boiler Construction Code.

3.6 Auxiliary equipment. -

3.6.1 Except as otherwise specified herein, all auxiliary equipment shall be furnished in accordance with the best commercial practice.

3.6.2 The automatic controls shall be as simple and rugged as practicable and shall be suitable for Marine service. Mercury tube actuated type controls are not acceptable.

3.6.3 Particular care shall be taken in design and construction of all control equipment to insure safety of the operating personnel from the danger of electric shock.

3.6.4 Oil burner. - The oil burner shall be of a mechanical pressure atomizing type, capable of smokeless operation when burning Diesel oil conforming to Specification MIL-F-16884.

3.6.5 The blower shall be designed to supply all requisite air to the boiler operating at its maximum rated capacity at a pressure 0.5 inch of water in excess of the pressure required to overcome all resistances of the burner and boiler.

3.6.5.1 Provisions shall be made for manual regulation of the amount of air supply to the burner as required for smokeless operation of the burner, also provisions shall be made for connecting air supply duct from the weather to the blower air intake.

3.6.6 Fuel oil pump. - The fuel oil pump shall be of a self-priming, continuous venting type suitable for installation in a two-pipe system. The pump shall have a minimum 12-foot suction lift.

3.6.7 Electric motors. - Electric motors for driven auxiliaries shall be in accordance with the following specifications:

Specification MIL-M-17556 for d.c. motors.
Specification MIL-M-17059 for a.c. motors.

3.6.7.1 The motors shall conform to the following classification requirements:

Horespower	As required.	
Voltage	As specified (for d.c. motors).	
Voltage, Number of phases, Frequency	As required (see 3.2 and 6.1)	for a.c.
Speed	Constant.	
Duty	Continuous.	
Service	C.	
Ambient temperature	50° C.	
Type	Squirrel cage (for a.c. motors).	
Winding	Shunt (for d.c. motors).	
Bearing	Ball.	
Enclosure	Dripproof protected.	

3.6.8 Rotary inverters. - Rotary inverters used to supply a.c. power for ignition shall be equivalent to service "C" as specified in Specification MIL-M-17556, and shall conform to the following classification requirements:

Size kilowatt (kw)	As required.
Voltage	110 volts.
Number of phases	Single.
Prime mover	D.c. motor.
Type	Revolving armature.
Duty	Continuous.
Ambient temperature	50° C.
Bearing	Ball.
Enclosure	Dripproof protected.

3.6.9 The automatic ignition and combustion control system shall be arranged for intermittent electric spark ignition and automatic "on and off" operation.

3.6.10 The thermostatic combustion control shall be of an immersion type. The control shall be arranged to start the burner when the temperature of the water in the boiler falls to a predetermined low limit, and to cut off the burner when the temperature of the water in the boiler rises to a predetermined limit. The control shall be suitable for setting the cut-in and cut-out temperature limits within a temperature range of 140° to 220° F., and shall have an adjustable differential permitting differential setting within a range of 12 to 25° F.

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3.6.11 The safety combustion control shall be arranged to stop burner operation within 90 seconds in the event of flame or ignition failure.

3.6.12 The fuel oil solenoid valve shall be installed in the fuel oil supply line to the burner, and shall be electrically wired to cut off the flow of fuel to the burner instantly when either thermostatic combustion control or the safety combustion control breaks electric current to stop the burner operation.

3.7 Identification plate. - Each boiler shall have an identification plate conforming to type A or B of Specification MIL-P-15024, except that identification plates of phenolic compounds are not acceptable. The identification plate shall contain the following information:

Name and description of article.
Serial number.
Federal stock number.
Name and address of manufacturer.
Contract number.
A square of about 3/4 inch with letters "INSP" at the top for affixing Government inspector stamp.
The words "U.S. PROPERTY".
Design pressure, p.s.i.
Operating pressure, p.s.i.
Hydrostatic test pressure, p.s.i.
Safety valve setting, p.s.i.
Electrical characteristics.

3.7.1 The stock number will be furnished to the contractor by the bureau or agency concerned on written request. Two blueprints of approved drawings showing boiler assembly views and containing boiler design data shall accompany the request.

3.8 Repair parts.

3.8.1 The repair parts shall be furnished as follows:

(a) Onboard repair parts.
(b) Stock repair parts.

3.8.2 Onboard repair parts shall be as specified in table II.

Table II - Onboard repair parts.

Part	Percent
Boiler and auxiliaries:	
Combustion chamber liner (if used).	100
Set of gaskets for boiler and burner.	800
Oil nozzles for burner.	600
Fuel pump complete.	100
Boiler and auxiliaries: (Cont'd)	
Pump (fuel) seal assembly.	200
Blower rotor bearings (set).	100
Drive belts (if used).	200
Flexible coupling of each size.	200
Oil relief valve.	100
Electrical:	
Safety combustion control.	100
Oil solenoid valve.	100
Oil solenoid valve coils.	400
Set of repair parts for DC motors and rotary inverter in accordance with Specification MIL-M-17556 plus one armature assembly.	100
Set of repair parts for AC motors in accordance with Specification MIL-M-17059.	100
Transformer, control circuit.	100
Transformer, ignition.	100
Ignition electrodes, set.	400
Ignition cables and connectors.	400

3.8.3 Stock repair parts shall be furnished in accordance with Specification MIL-P-15137.

3.8.4 "Per ship" stock repair parts, when required, shall be furnished as specified (see 6.1).

3.8.5 "Per country" stock repair parts, when required, shall be furnished as specified (see 6.1).

3.8.6 Marking for identification. - Identification marking of repair parts shall be in accordance with Standard MIL-STD-130.

3.9 Special tools and wrenches. - The set of special tools and wrenches, except as otherwise specified herein, shall be furnished as specified (see 6.1) in accordance with the contractor's recommendations and shall consist of special tools and wrenches necessary for proper maintenance of boiler and auxiliaries. No ordinary tool shall be included. Each tool shall be indelibly marked to indicate the service for which intended. Special tools are defined as those tools not listed in the Navy Stock List of General Stores. (Copies of this list may be consulted in the office of the Government inspector.) The set shall include the following:

2 - tube brushes.
1 - tube brush handle.

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3.10 Drawings. - Drawings and lists shall be furnished as required herein and in accordance with Specification MIL-D-70327 amplified as follows:

Class	2
Type	As specified herein
Tracing material	Cloth
Reproducibles	One copy of each drawing sheet, photographic, type I and II as required, class 2 of Specification MIL-D-5480
Blueprints	Three copies of each drawing sheet (blueprint or blue line prints); type I and II as required, class 1 of Specification MIL-D-5480.

3.10.1 Drawings shall be submitted to the bureau or agency concerned within thirty days of the date of contract. Submittal, review, approval and validating procedure and requirements shall be in accordance with Specification MIL-D-963. The use of manufacturer's standard drawings together with certification data sheets to show design variations, material, or other changes is encouraged. Drawings shall contain views, details, data and tabulations as specified for drawings entitled as follows:

(a) Assembly views (front, rear, and side elevations), plan view, and foundation of the boiler complete with all auxiliaries, accessories, and controls. All components of the assembly shall be identified by piece numbers and listed in the list of material. Overall dimensions, locations, type and size of all terminal connections, and space required for tube renewal shall be shown. The following data shall be tabulated:
  (1) Pressure, designed, psi
  (2) Pressure, operating, psi
  (3) Pressure, safety relief valve setting, psi
  (4) Pressure, hydrostatic test, psi
  (5) Heating surface, square feet
  (6) Furnace volume, cubic feet
  (7) Boiler capacity, B.t.u./hour
  (8) Oil rate, pounds/hour for full capacity operation.
  (9) Weight, dry, pounds
  (10) Weight, wet, pounds

(b) Sectional views of the boiler and burner assembly. This drawing shall include arrangement of and thickness of pressure parts, method of tube attachment to tube sheets, arrangement of internal fittings, details and identification of refractory and insulation used, and stress calculations for the pressure vessel.

(c) All details of the boiler. This drawing shall include details of welds, specification for electrodes and descriptions of welding procedure, and heat treatment to be used in fabrication of the pressure vessel.

(d) All details of the burner.

(e) Dimensional assembly and sectional view for all auxiliaries, accessories, and controls with all parts shown, identified, and listed in the list of material. These drawings shall describe material, rating and manufacturer's model number; size type, and location for all connections mounting details, and weight of the complete component.

(f) Drawings for electric motors in accordance with Specifications MIL-M-17556 and MIL-M-17059.

(g) Diagrammatic equipment wiring and elementary schematic wiring diagrams. This drawing shall include step-by-step description of operation of the electrical control circuit and safety devices.

3.10.2 All work done prior to approval of drawings shall be entirely at the contractor's risk.

3.11 Technical manuals. - Technical manuals shall be type II in accordance with Specification MIL-M-15071. Content of the manual shall be arranged in accordance with Specification MIL-M-15071, except that parts identification is not required. In lieu thereof, a "List of Components" comprising major parts of the boiler and burner assemblies and all individual auxiliaries, accessories, and controls furnished with the boiler, shall be included. The "List of Components" shall be arranged as follows:

Column 1 - Assembly piece number.
Column 2 - Manufacturer's part number.
Column 3 - Quantity furnished.
Column 4 - Name, in full (abbreviations are not desired).
Column 5 - Description containing application or function, material, size, rating, manufacturer's type or model number, and catalog number.
Column 6 - Federal stock number (to be added when available).
Column 7 - Manufacturer's or contractor's drawing number.
Column 8 - Bureau drawing number.
Column 9 - Weight.

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3.11.1 The list of repair parts and the list of tools and wrenches furnished shall be included in the manual. The arrangement used for these lists shall be the same as specified for the "List of Components" specified in 3.11.

3.11.2 Reduced size reproductions of the following drawings shall be included in the manual:

(a) Assembly views of the boiler, plan view and foundation.
(b) Sectional views of the boiler and burner assembly.
(c) Assembly and sectional views of all auxiliaries.
(d) Diagrammatic equipment wiring and elementary schematic wiring diagrams.
(e) Master drawings for motors.

3.11.3 Manufacturer's standard bulletins applicable to auxiliaries, accessories, and controls furnished shall be included in the manual where necessary for complete description of construction, method of installation, operation, adjustment, and maintenance.

3.11.4 The method of approval of the manual shall be as specified in Specification MIL-M-15071, except that two sample copies of the manual, complete, shall be submitted for approval to the bureau or agency concerned.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept compete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. - Preproduction inspection shall be conducted in accordance with 4.2.1 unless otherwise specified by the bureau or agency concerned.

4.2.1 Preproduction inspection shall be conducted as follows to determine conformance with the requirements of this specification and to ascertain fulfillment of the capacity and efficiency requirements and satisfactoriness of control features and safety devices: The test inspection shall be prepared by the laboratory designated to make the test and submitted to the bureau or agency concerned for approval prior to test.

(a) Weight of the boiler dry and wet.
(b) Hydrostatic tests before and after the performance test run.
(c) Operational tests to determine correctness of adjustment, effectiveness, range, and reliability of control and safety devices.
(d) Performance test consisting of at least a 2 hour run to determine and record the following:
  (1) Fuel consumption.
  (2) Heating capacity.
  (3) Thermal efficiency.
  (4) Temperature, pressure, and CO_2 content of flue gases.
  (5) Electrical power input.
  (6) Air, oil, and water pressure.
(e) Radio interference test to determine compliance with suppression requirement of Specification MIL-I-16910.

4.2.2 Unless otherwise specified by the bureau or agency concerned, boilers submitted for preproduction test shall be size 2 designed for operation with 115 volts d.c.

4.3 Examination and tests at place of manufacture. - Each boiler offered for delivery shall be inspected as follows:

(a) To determine compliance with the approved drawings.
(b) Hydrostatic test.
(c) To determine compliance with preservation, packaging, and packing requirements.
(d) Power input to electrical apparatus.
(e) Insulation resistance.
(f) Operation of safety devices.
(g) Weight of the boiler dry and wet.

4.4 Additional inspection. - Where other specifications form a part of this specification, sampling, inspection, and tests shall be conducted as required in reference specifications.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of shipboard repair parts. -

5.1.1 Boiler. - The boiler shall be shipped assembled as a complete unit, except such accessories which require separate packing for proper preservation and security in shipment.

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford

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adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation of 30 days. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts and tools. - Onboard repair parts and tools shall be processed as follows:

(a) Mechanical repair parts and tools shall be preserved and packaged level A, packed level C and marked levels A and C respectively in accordance with the requirements of Specification MIL-R-196 and MIL-P-17286 as applicable.
(b) Electrical repair parts shall be preserved and packaged level A, packed level C and marked levels A and C respectively in accordance with Specification MIL-P-16298 and MIL-E-17555 as applicable.

5.1.2.1 Repair parts boxes. - When onboard repair parts and tools are destined for a ship that is not fitted with bin or drawer type stowage (see 6.1), shipboard repair parts shall be packed in repair parts boxes conforming to Specification MIL-B-233 and the following:

(a) Class I - carbon steel construction. - Repair parts boxes shall be type M or W, as specified (see 6.1).
(b) Class II - nonmagnetic construction. - Repair parts boxes shall be type W, except that all hardware on the boxes shall be made of nonmagnetic material (see 3.3.3.1).

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1).

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Level A. -

5.2.1.1.1.1 The boiler and accessories shall be cleaned, preserved, and packaged in accordance with Specification MIL-B-3180, as applicable.

5.2.1.1.1.2 The accessories which are shipped knockdown (see 5.2.1.2.1) and repair parts and tools shall be preserved and packaged as follows:

(a) Mechanical parts and tools shall be preserved and packaged in accordance with the level A requirements of Specification MIL-R-196 and MIL-P-17286 as applicable.
(b) Electrical parts shall be preserved and packaged in accordance with the level A requirements of Specifications MIL-P-16298 and MIL-E-17555, as applicable.

5.2.1.1.2 Level C. - Preservation and packaging of basic equipment shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation (30 days).

5.2.1.2 Packing. -

5.2.1.2.1 General requirements for levels A and B. - The boiler shall be shipped assembled as a complete unit, except such accessories which require separate packing for proper preservation and security in shipment. All containers shall be new and shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking. Methods of cushioning and materials required to protect items from damage due to shock and vibration shall be in accordance with Specification JAN-P-100 and the appendix to Specification MIL-C-104.

5.2.1.2.2 Equipment. -

5.2.1.2.2.1 Level A. - Boilers shall be individually packed in sheathed crates conforming to Specification MIL-C-104. Accessories packed as specified in 5.2.1.2.2.1.1 shall be secured within the shipping container of the basic equipment. Crate closure shall be in accordance with the appendix to Specification MIL-C-104.

5.2.1.2.2.1.1 Detached accessories shall be packed in cleated plywood, nailed wood or fiber boxes

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conforming to Specifications PPP-B-601, PPP-B-621 (class 2) and PPP-B-636 (class 2), respectively. Boxes shall be lined with a sealed case liner conforming to Specification MIL-L-10547. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds unless the weight of a single item exceeds that weight. Box closure shall be in accordance with the applicable box specification or appendix thereto.

5.2.1.2.2.2 Level B. - Boilers shall be individually packed in sheathed crates conforming to Specification MIL-C-104. Accessories packed as specified in 5.2.1.2.2.2.1 shall be secured within the shipping container of the basic equipment.

5.2.1.2.2.2.1 Detached accessories shall be packed in snug-fitting cleated plywood, nailed wood, corrugated or solid fiber boxes conforming to Specification PPP-B-601, PPP-B-621 (class 1) or PPP-B-636 (class 1), respectively. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds unless the weight of a single item exceeds that weight. Box closure shall be as specified in the applicable box specification or appendix thereto.

5.2.1.2.3 Repair parts. -

5.2.1.2.3.1 Stock repair parts. - Stock repair parts shall be snugly packed in shipping containers as specified in 5.2.1.2.2.1.1 or 5.2.1.2.2.2.1 as specified.

5.2.1.2.3.2 "Per ship" and "per country" stock repair parts. - "Per ship" and "per country" stock repair parts shall be snugly packed in shipping containers as specified in 5.2.1.2.2.1.1.

5.2.1.3 Technical publications. - Technical manuals, pamphlets, and handbooks, as applicable, shall be packaged and packed for the level specified in accordance with Specification MIL-M-15071.

5.2.1.4 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129 and in addition, shall include the electrical and mechanical operational characteristics or rating (as applicable).)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class, style, and size of the boiler (see 1.2 and 3.5.7).
(c) Electric power supply available for operation of the boiler and auxiliaries (see 3.2 and 3.6.7.1).
(d) Number of sets and list of "per ship" stock repair parts (see 3.8.4).
(e) Number of sets and list of "per country" stock repair parts (see 3.8.5).
(f) Number of sets of special tools and wrenches (see 3.9).
(g) Whether or not the ships for which onboard repair parts and tools are destined are equipped with bin or drawer type stowage and if not specify type of repair parts boxes required (see 5.1.2.1).
(h) Preservation packaging, packing or marking requirements other than those required by paragraph 5.1(see 5.2).

6.2 Bid data. - Bidders shall furnish with their bids the following information in duplicate:

(a) Rated capacity B.t.u./hr.
(b) Heating surface square feet.
(c) Furnace volume cubic feet.
(d) Designed efficiency.
(e) Quantity of oil in pounds per hour required for operation at rated capacity.
(f) Complete description of the boiler, burner, and auxiliaries.
(g) Complete description of controls (manufacturer's bulletins shall be attached).
(h) Weights of the boiler dry and wet, pounds.
(i) Itemized list of onboard repair parts with unit price for each item.
(j) Itemized list of tools and wrenches, with unit price for each item.
(k) Estimated weight of repair parts and tools and wrenches.
(l) Two sets of blueprints of the following preliminary drawings and data to describe adequately the equipment the bidder proposes to furnish:

(1) Assembly views (front, side, rear, and plan) of the boiler, complete with all auxiliaries, accessories and controls, giving overall dimensions, space required for tube renewal, and locations, type and size of all connections. All components of the boiler assembly shall be identified by piece numbers and listed in the list of material.
(2) Sectional elevations of the boiler and burner assembly showing arrangement of tubes and external parts and giving sizes of tubes, thicknesses of shell plates, and identifying refractory and insulation used.

If a bidder proposes to furnish equipment which previously has been supplied and for which

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the bureau or agency concerned has approved drawings on record, reference to the approved drawings will be acceptable in lieu of preliminary drawings.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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